

PROJECT BASED EXPERIMENTAL LEARNING

COURSE – NOSQL

PROJECT REPORT

TOPIC – LIBRARY MANAGEMENT SYSTEM USING MONGODB



SUBMITTED TO

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REPORT BASIS

This report presents the practical work carried out for the NoSQL course through a Library Management System built on MongoDB, a document-oriented NoSQL database.

ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher Mr. Harendra Singh for giving me the opportunity to work on this NoSQL project titled “Library Management System Using MongoDB”.

This project helped me understand the basic concepts of NoSQL databases and learn how MongoDB is used to create, retrieve, update, sort and delete data. I also learned how different query operators such as greater than, less than, equal to, AND, and OR are used in MongoDB.

I am thankful to my teacher for the guidance and support provided throughout the completion of this project.

I also acknowledge the learning resources and tools used during the implementation of the project, which helped me practice MongoDB queries and understand document-based data management.

— ARNAV MISHRA

ABSTRACT

This report presents a Library Management System developed using MongoDB, a NoSQL document-oriented database. The project demonstrates how library book information can be stored as documents and managed through MongoDB queries and operations.

The system uses a Library database and a books collection containing 50 book records. Each record includes fields such as book ID, title, category, publication year, and number of copies available. The project demonstrates database creation, collection creation, insertion, retrieval, filtering, updating, deletion, sorting, limiting, category-based searching, and document counting.

Comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte and logical operators such as \$and and \$or are also demonstrated. The project therefore provides practical exposure to core NoSQL database operations using MongoDB.

KEY FEATURES

- Document-based book data storage using MongoDB.
- Creation of a database and collection.
- Insertion of 50 book documents.
- CRUD operations: create, read, update and delete.
- Comparison and logical query operators.
- Sorting, limiting, category-based searching and counting.
- Practical execution and observation of MongoDB query outputs.

INTRODUCTION

MongoDB is a NoSQL database management system that stores information in document form. Instead of organizing data only in fixed rows and columns, MongoDB stores records as flexible documents. This makes it suitable for applications where data needs to be stored and queried in a document-oriented structure.

In this project, a Library Management System has been created using MongoDB. The database contains information about 50 books, including book ID, title, category, publication year and copies available. A series of MongoDB operations are performed on these records to demonstrate practical database management.

The project also demonstrates how records can be filtered using comparison operators, combined using logical operators, modified using update operations, removed using delete operations, ordered using sort, restricted using limit, searched by category and counted.

OBJECTIVES

- To understand the concept of NoSQL databases.
- To learn the basic features and document model of MongoDB.
- To create a database and collection in MongoDB.
- To insert and manage 50 book records.
- To perform CRUD operations on book documents.
- To use comparison operators such as \$eq, \$gt, \$lt, \$gte and \$lte.
- To perform logical AND and OR operations.
- To sort, limit, search and count documents.

PROJECT DATA MODEL

FIELD	EXAMPLE	PURPOSE
bookid	101	Unique book ID
title	Introduction to Algorithms	Title of the book
category	Computer Science	Subject category of the book
year	2009	Year of publication
copies	8	Number of copies available in the library

METHODOLOGY

The project follows a practical sequence of MongoDB operations. The database is first selected, followed by collection creation and insertion of the book dataset. The stored records are then queried and manipulated using MongoDB operators.

STEP	ACTIVITY	DESCRIPTION
1	Create Database	Create or switch to the Library database using use Library.
2	Create Collection	Create a books collection inside the Library database.
3	Insert Records	Insert 50 book documents containing book ID, title, category, year and copies.
4	Retrieve Data	Display the stored book documents using find().
5	Filter Data	Use comparison operators to retrieve records based on publication year.
6	Update Data	Use updateOne() and updateMany() to modify documents.
7	Delete Data	Use deleteOne() and deleteMany() to remove records.
8	Logical Queries	Use \$and and \$or to combine conditions.
9	Organize Results	Use sort() and limit() to control query results.
10	Search and Count	Search by category and count documents in the collection.

BASIC DATABASE SETUP

```
use Library
```

This creates or switches to the Library database.

```
db.createCollection("books")
```

This creates the books collection.

IMPLEMENTATION DETAILS

The implementation is based on the MongoDB shell/query syntax. The book dataset contains 50 documents with book ID, title, category, year and copies.

INSERTING BOOK RECORDS

```
db.books.insertMany([
  {bookid:101, title:"Introduction to Algorithms", category:"Computer Science", year:2009, copies:8},
  {bookid:102, title:"Clean Code", category:"Computer Science", year:2008, copies:6},
  {bookid:103, title:"The C Programming Language", category:"Computer Science", year:1978, copies:5},
  ...
  {bookid:150, title:"Mindset", category:"Self Help", year:2006, copies:6}
])
```

All 50 book documents are inserted successfully into the books collection.

DISPLAYING RECORDS

```
db.books.find()
```

The find() operation retrieves documents from the books collection. The shell displays the records in batches of 20 documents.

COMPARISON OPERATORS

OPERATOR	EXAMPLE QUERY	PURPOSE
\$eq	db.books.find({ year: { \$eq: 1997 } })	Published exactly in 1997
\$gt	db.books.find({ year: { \$gt: 2018 } })	Published after 2018
\$lt	db.books.find({ year: { \$lt: 1950 } })	Published before 1950
\$gte	db.books.find({ year: { \$gte: 2018 } })	Published in 2018 or later
\$lte	db.books.find({ year: { \$lte: 1960 } })	Published in 1960 or earlier

MONGODB OPERATIONS DEMONSTRATED

UPDATE ONE

```
db.books.updateOne({ bookid: 102 }, { $inc: { copies: 1 } })
```

Increases the available copies of "Clean Code" by one. The output shows matchedCount: 1 and modifiedCount: 1.

UPDATE MANY

```
db.books.updateMany({ category: "Computer Science" }, { $inc: { copies: 5 } })
```

Adds 5 copies for every Computer Science book. The output shows 12 documents matched and modified.

DELETE ONE

```
db.books.deleteOne({ bookid: 101 })
```

Deletes the single document with book ID 101.

DELETE MANY

```
db.books.deleteMany({ category: "History" })
```

Deletes all History-category documents; 10 documents are deleted.

AND

```
db.books.find({ $and: [ { year: { $gte: 2000 } }, { category: "Science" } ] })
```

Returns Science books published in the year 2000 or later (2 result).

OR

```
db.books.find({ $or: [ { year: { $lte: 1950 } }, { category: "Self Help" } ] })
```

Returns books published in 1950 or earlier, or books in the Self Help category (14 results).

SORT

```
db.books.find().sort({ year: 1 }).limit(5)
```

Sorts records by publication year in ascending order; descending order is demonstrated with -1.

LIMIT

```
db.books.find().limit(3)
```

Displays only the first three remaining documents.

SEARCH BY CATEGORY

```
db.books.find({ category: "Science" })
```

Finds books whose category is Science (9 results).

COUNT

```
db.books.countDocuments({})
```

Counts documents currently present in the collection; the final result is 39.

RESULTS

The project successfully demonstrated the core NoSQL operations required for managing a book dataset in MongoDB. The original dataset contained 50 book documents, and the queries were executed to retrieve, filter, modify, delete and organize the records.

OPERATION	RESULT	OBSERVATION
Initial records inserted	50	50 book documents inserted successfully.
updateOne()	1 matched / 1 modified	Copies of book ID 102 increased by one.
updateMany()	12 matched / 12 modified	Copies increased by 5 for Computer Science books.
deleteOne()	1 deleted	Book ID 101 removed.
deleteMany()	10 deleted	History-category documents removed.
Final count	39	Count reported after the deletion operations.

LEARNING OUTCOMES

- Understanding of the MongoDB document model and NoSQL approach.
- Practical use of database and collection creation commands.
- Hands-on experience with CRUD operations.
- Understanding of comparison and logical operators.
- Ability to sort, limit, search and count MongoDB documents.
- Understanding of how query results change after update and delete operations.

PROJECT RESULT

The project demonstrates a complete practical workflow for managing a library book dataset in MongoDB, from database creation and insertion through querying, updating, deletion and final document counting.

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Library Management System containing book information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update and delete documents. I also learned how to sort data in ascending and descending order, how comparison operators such as \$eq, \$gt, \$lt, \$gte and \$lte work, and how logical operators such as \$and and \$or can be used to combine conditions.

Overall, this project helped me understand how MongoDB can be used to manage and query library data efficiently.

FUTURE SCOPE

- Develop a web-based library interface connected to MongoDB.
- Add authentication and role-based access for librarians and members.
- Add member records, book issue/return tracking and fine calculation modules.
- Add dashboards and charts for book circulation and popularity analytics.
- Deploy the database using MongoDB Atlas and the application on a cloud platform.
- Add validation, indexing and more advanced aggregation queries for larger datasets.

REFERENCES

- MongoDB Documentation – MongoDB Query and CRUD concepts.
- MongoDB documentation on comparison and logical query operators.
- MongoDB documentation on update, delete, sort, limit and count operations.

THANK YOU!

Library Management System Using MongoDB

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